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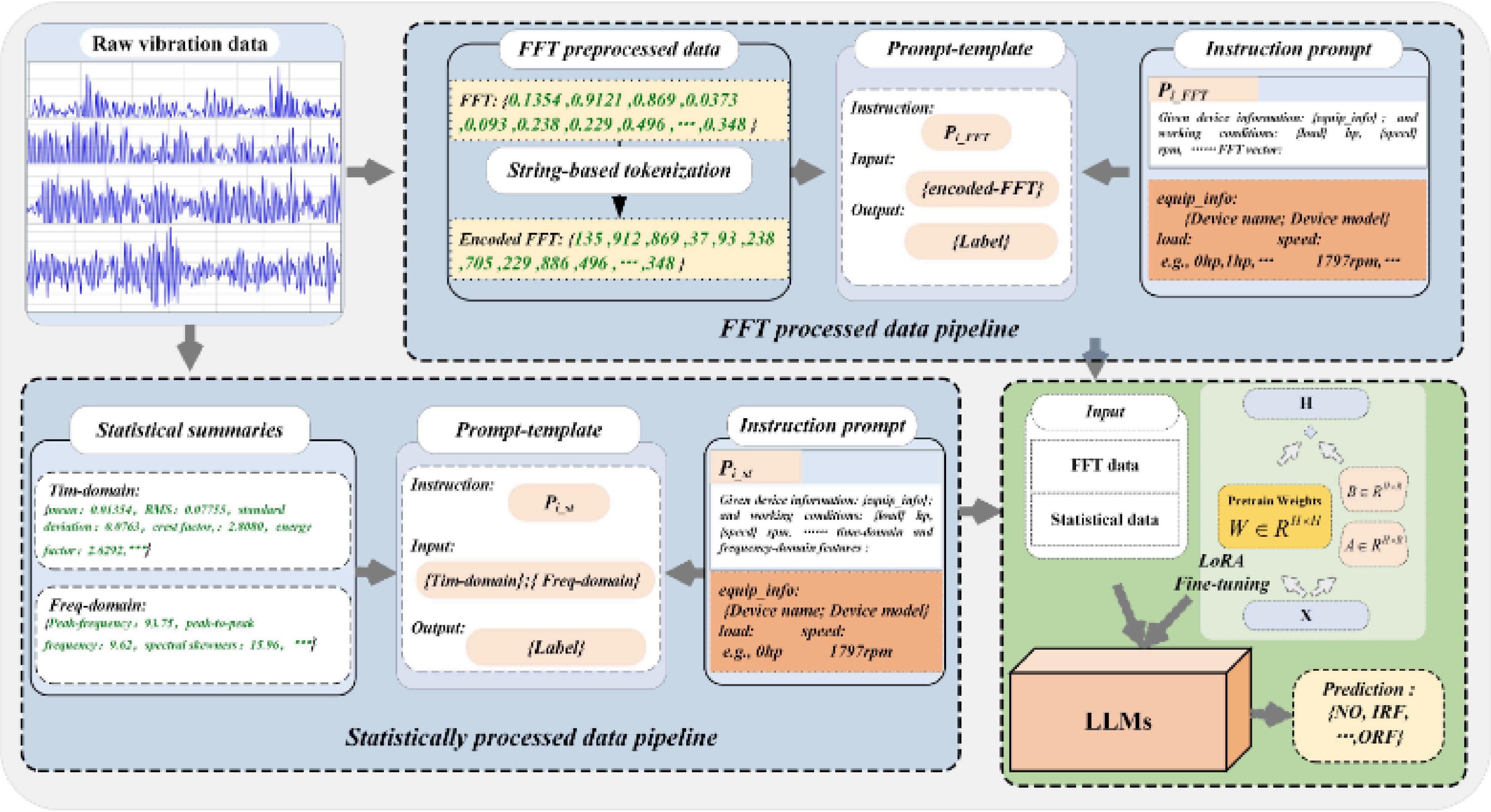
FD-LLM

Project Report

https://github.com/Bhavy-Ranka/llm-fault-diagnosis.git

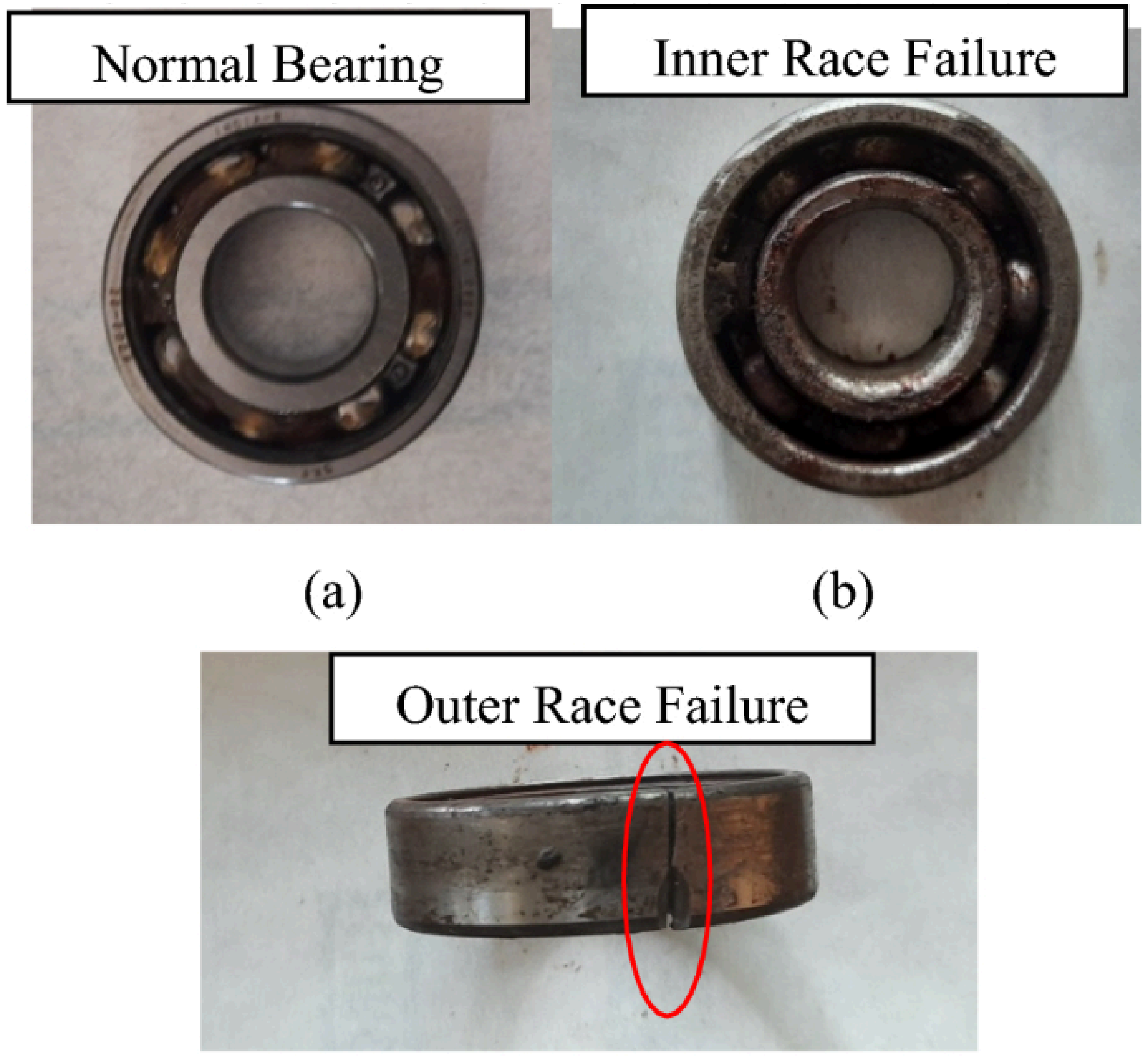
02 Research Paper & Dataset

- **Research Paper :** FD-LLM: LARGE LANGUAGE MODEL FOR FAULT DIAGNOSIS OF MACHINES (Link)
- **Dataset :** CRWU Bearing dataset (Link)
- **Custom Dataset :** Statistics feature and FFT custom json files.(Link)



Paper Architecture

03 Major Components



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- **Data Visualization** —Analyzed bearing vibration data by filtering its frequency spectrum and then examining it through envelope analysis, a spectrogram, kurtosis, and PSD to detect defect frequencies.

- **Data Preprocessing & Feature Engineering Pipeline** — Raw CWRU vibration signals are segmented into fixed-size windows and converted into FFT and statistical (time + frequency domain) feature sets, then packaged into labeled instruction prompts.

- **Fine-Tuning the Llama Model on Statistical Data** — The Llama model is fine-tuned (e.g., via LoRA) on the statistical-feature instruction dataset to map time/frequency-domain features and operating conditions to bearing fault labels.

- **Evaluation of the Fine-Tuned Model** — The fine-tuned model's predicted fault labels are compared against ground truth on a held-out test set using accuracy, precision/recall, and confusion-matrix analysis.

Data Preprocessing & Feature Engineering Pipeline

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- Segment: Raw bearing signals split into 2048-sample windows

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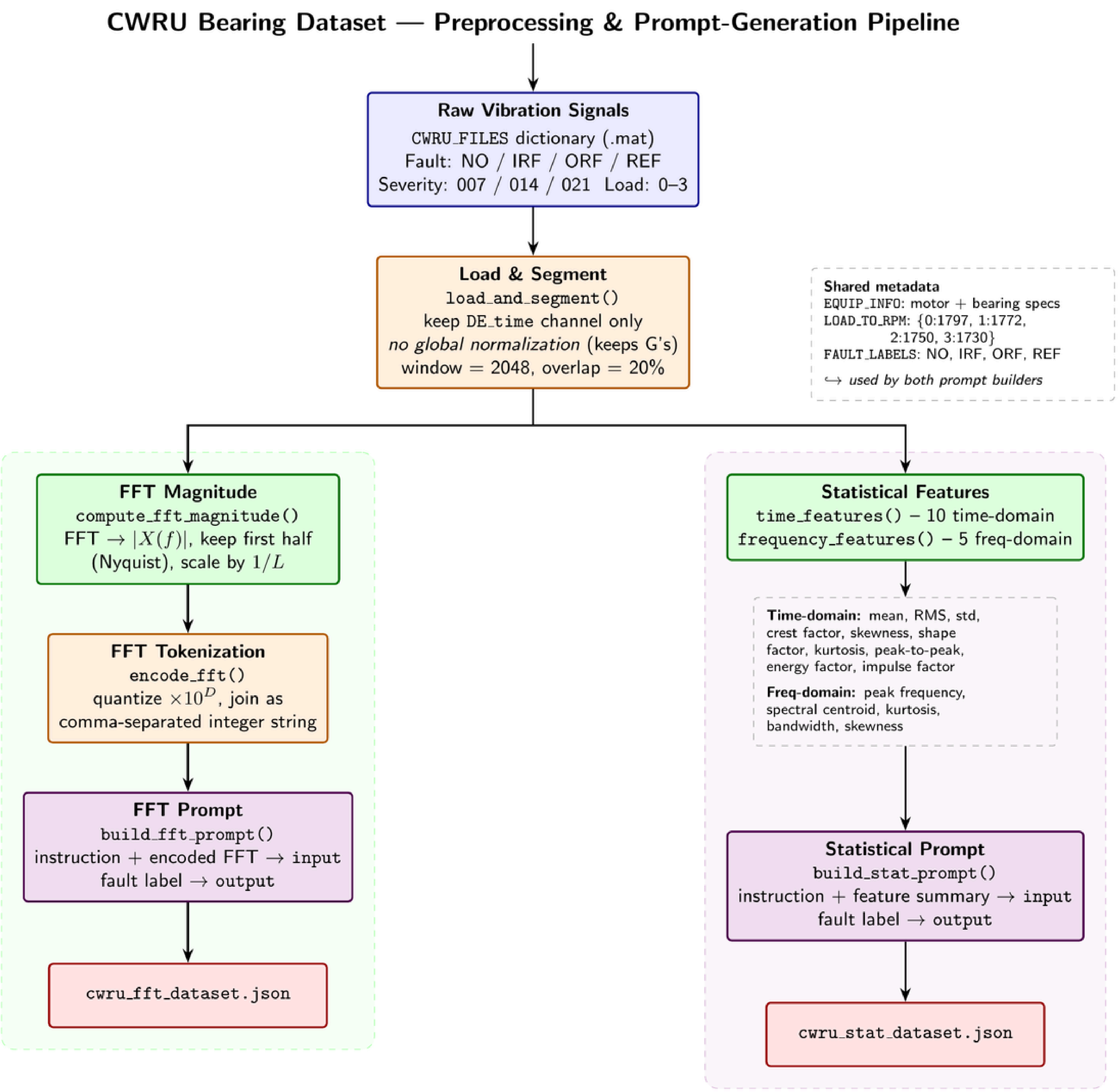
- Extract Features: Each segment → FFT spectrum + 15 stats (time + freq domain).

- Build Prompts: Features + labels packaged into instruction-style JSON datasets for fine-tuning.



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Prompt Template

Dataset	Prompt-template
FFT processed samples	"Instruction": Given machine information: {equip_info}; and working conditions: {load} hp, {speed} rpm, please predict the operating status of the bearing based on the following FFT vector. "Input": {FFT} "Output": {label}
Statistically processed samples	"Instruction": Given machine information: {equip_info}; and working conditions: {load} hp, {speed} rpm, please predict the operating status of the bearing based on the following time-domain and frequency-domain features. "Input": {Tim-domain features}; {freq-domain features} "Output": {label}

Prompt template of research paper


```
"instruction" :
string "Given machine information: CWRU Bearing Dataset; Device: Reliance Electric 2 hp motor; Bearing model: SKF deep-groove ball bearing 6205-2RS JEM; Pitch diameter: 1.748 in; Ball diameter: 0.3126 in; Contact angle: 0°; Number of balls: 9; and working conditions: 0 hp, 1797 rpm, please predict the operating status of the bearing based on the following time-domain and frequency-domain features."

"input" :
string "mean: -0.0436, RMS: 1.0199, standard_deviation: 1.0189, crest_factor: 3.0145, skewness: -0.0949, shape_factor: 1.2393, kurtosis: -0.3367, peak_to_peak: 5.8348, energy_factor: 0.0007, impulse_factor: 3.7358, peak_frequency: 1037.1094, peak_to_peak_frequency: 5994.1406, spectral_kurtosis: 142.2988, spectral_bandwidth: 2188.7544, spectral_skewness: 9.8495"
"output" : string "NO"
}
```


Implemented Prompt template

Fine-Tuning the Llama Model on Statistical Data

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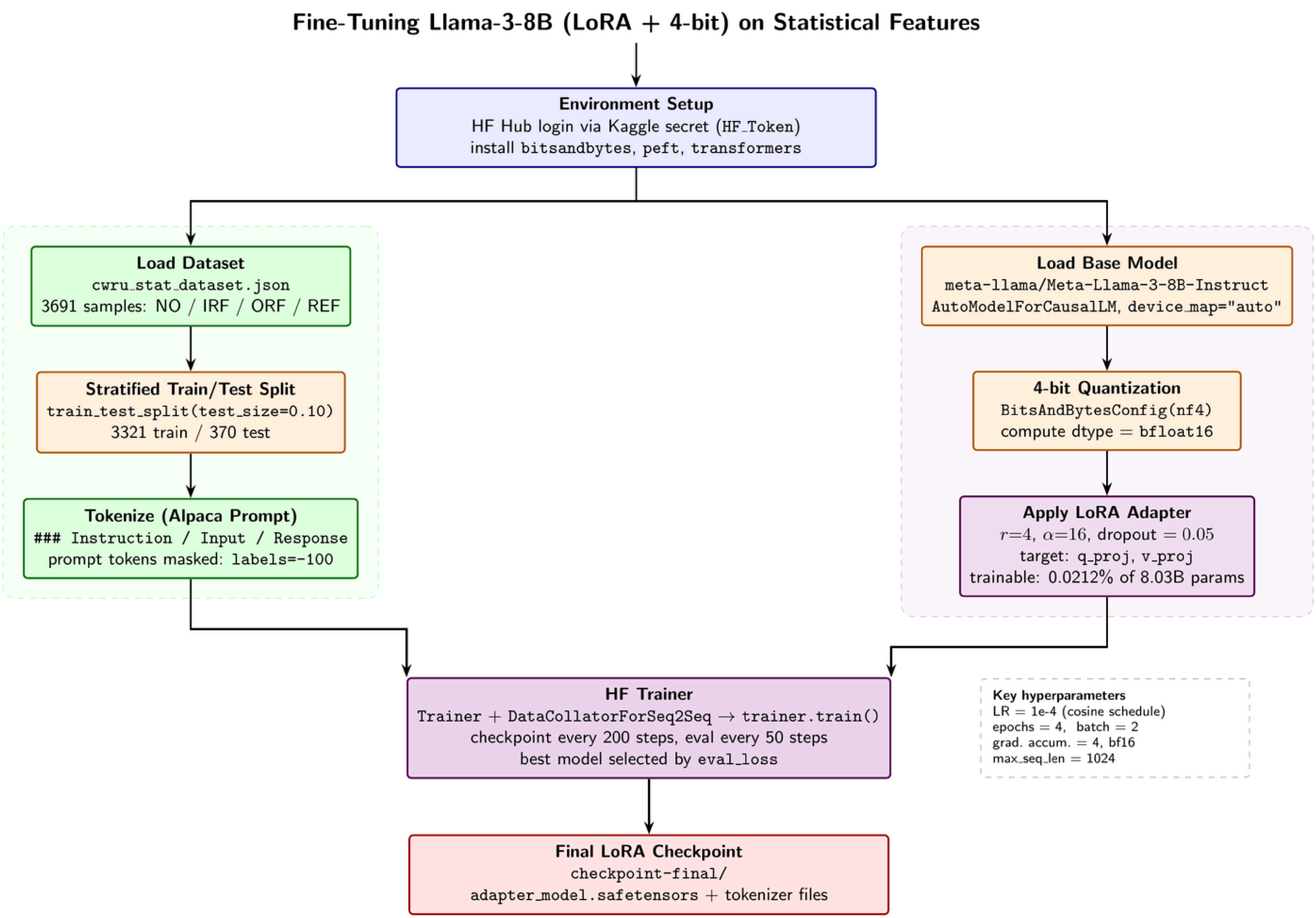
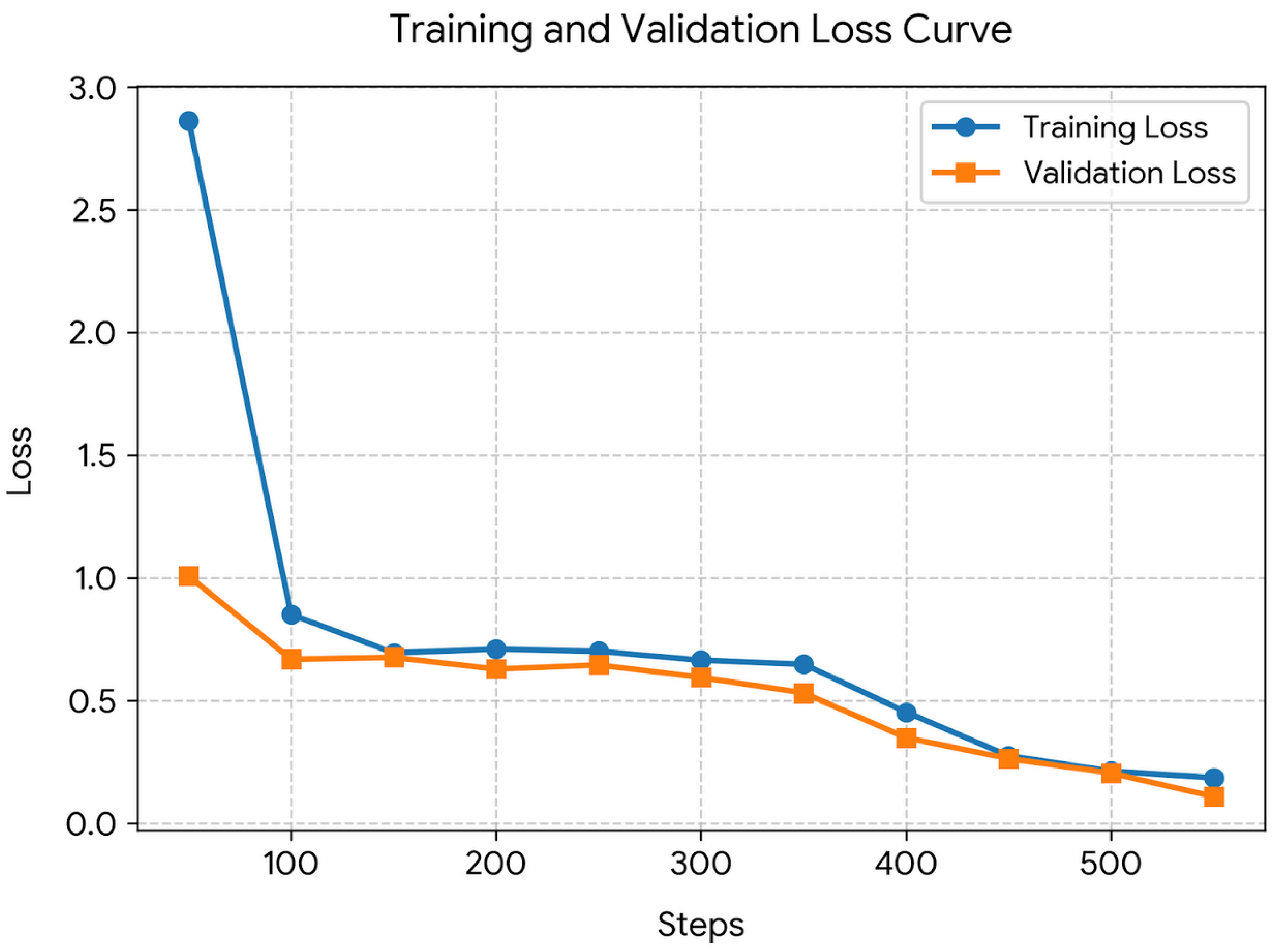
- Parallel Prep: Dataset is loaded/split/tokenized while Llama-3-8B is loaded, 4-bit quantized, and fitted with a LoRA adapter (only 0.02% params trainable).
- Train & Save: Both branches feed into HF Trainer, which fine-tunes the model and saves the final LoRA checkpoint.



Training loss and Validation loss



Evaluation of the Fine-Tuned Model

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- Setup: Reload the held-out test split and reattach the trained LoRA adapter to the quantized base model.
- Inference: Generate predictions on test prompts (no output given) and extract fault labels from the model's response.
- Results: Compute accuracy/precision/recall/F1 and confusion matrix — achieved 92.16% accuracy (F1 = 0.921) across NO/IRF/ORF/REF classes.